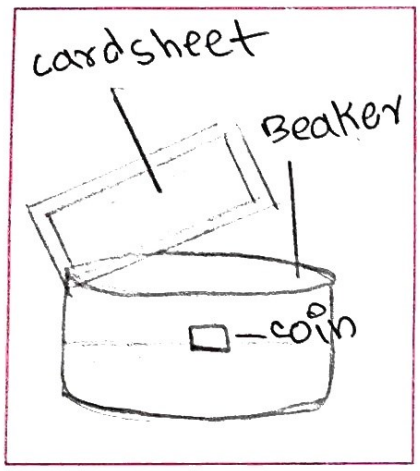
# Experiment No.1

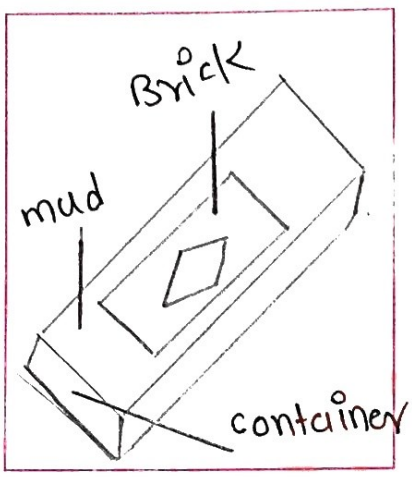
**Title: To Study Newton's Laws of Motion.**

**Apparatus :** Beaker, Piece of Card sheet, Sand, Metal Coin, Metal/Wooden Blocks of equal size, Soil, Water, Flat Surface Pot, Balloon, Piece of rubber Pipe (Diameter about 5mm to 1 cm)

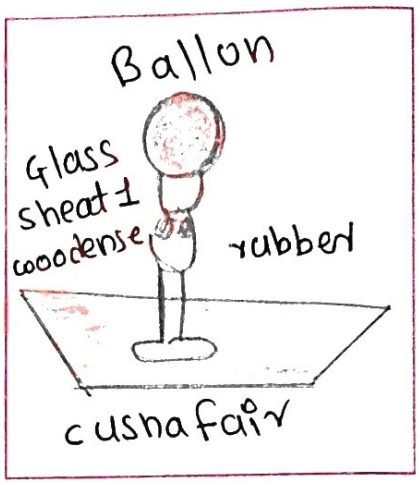
**Figure :** Draw labelled figures for three procedures.




A




B




C

**Procedure A :**

1. Take some sand in beaker.
2. Place a cardsheet on it and keep a metal coin in the centre.
3. Now strike a cardsheet with a flick of the finger.

**Procedure B :**

1. Take some soil in a flat container. Mix it with water and keep the container near the legs of the table.
2. Push the wooden and metal block of same size in such a way that they fall in the flat container.
3. Observe the impressions created by the blocks in the soil mixture in the container.

**Procedure C :**

1. Tie a balloon tightly to a light weighted rubber tube.
2. Inflate the balloon by blowing through the rubber tube and tightly close the end of the rubber tube by pressing with finger.
3. Without detaching the finger, place the balloon on a flat surface and then remove the finger.

**Observation : Observe the three procedures carefully and notedown the observations .**

1. The cardsheet on it and keep a metal coin it centre. Now strike a cardsheet with a flick of finger.
2. The coin does not moves with the cardsheet but falls due to gravity as the earth.
3. These observation are based on newtons first low of motion.

Inference :

1. The experiment is law of interia elst law of newlon's means any object remain steet.
2. This on 2nd ~~netoon~~ newton law of motion of momentum it is product of mass and velocity.
3. These is based on 3rd netotn law of motion means every action force equal & opposite reaction force which is simuten-elour.

## Multiple Choice Questions

1. Newton's ..... law of motion is used in rocket launching.
   A. First ✓B. Third C. Second D. Gravitational
2. While doing the procedure 1, the coin fell in the beaker due to the property of the coin called.....
   A. Momentum B. Weight ✓C. Inertia D. Force
3. The forces of action and reaction have the same .....and opposit .....
   ✓A. Unit, Direction B. Direction,Unit C. Momentum, Inertia D.Inertia, Momentum
4. On release of a bullet from the gun, the gun moves to back slightly . This motion is called as .....
   A. Impact B. Imbalanced Force C. Inertia ✓D. Recoil
5. Momentum = .....
   A. $\frac{\text{Velocity}}{\text{Mass}}$ B. $\frac{\text{Acceleration}}{\text{Time}}$ ✓C. Velocity x Mass D. Speed x Mass

## Questions

1. Which other procedures will you suggest to study Newton's Laws of motion in the Laboratory?

Newton's second law of motion take two object have different massel push those object all together then it show that hevier mass object get great force and slower mass get little force

2. Explain some examples from our everyday life based on Newton's laws of motion.

1) When we more by bicycle.
2) Newton's 3rd law when we just forward, the skale's go back coord.
3) When we hit any also hit by same force.

3. Explain Newton's laws of motion with the help of one single example in daily life .

Newton's 1st laws at motion Rotating fan the fan rotating continusly init's motion when switch off an electricity borad if rotating of the same time without electricity and intertia

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